

Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

grn_tf_activity_model.py

This builds the gene regulatory network. Rather than listing which genes changed, it asks which transcription factors, the master switches that control other genes, are behind those changes. The logic is that if a TF is switched off, the genes it normally drives will drop together, so you can infer the TF's activity from the behaviour of its targets. It uses the CollecTRI regulons and decoupler, and because those are human resources the zebrafish genes have to be mapped to human ones first.

- **orthology_map_zfish_to_human.csv:** the zebrafish to human gene dictionary from g:Profiler. It is saved to disk so later runs do not have to fetch it again, and several of the later scripts reuse this same file so that everything stays consistent. Note the mapping is lossy, since not every zebrafish gene has a clean human equivalent.
- **tf_activity_Q1.csv:** every transcription factor that could be scored, with an activity score and a p-value, ranked. A positive score means the TF looks switched on in the mutant, a negative score means switched off. Out of 698 TFs scored, 71 are significant: 63 activated and 8 repressed. The 8 repressed ones matter far more than their number suggests, because they include the whole E2F family, which runs cell division.
- **tf_activity_barplot_Q1.png:** the most activated and most repressed TFs drawn as a bar chart, sorted by score. This is the figure where the E2F family stands out clearly on the repressed side and the stress and p53 factors on the activated side.
- **grn_model_edges.csv:** the network itself, written as an edge list. Each row is one regulatory relationship: which TF, which target gene, and whether the TF activates or represses that target. There are 3,352 such edges linking 71 TFs to 965 target genes. Being a plain table means it can be reused by later scripts, which is exactly what the mapping script does.
- **grn_model.graphml:** the same network in GraphML format, which is the format Cytoscape reads, so the network can actually be drawn and explored visually rather than only read as a table.
- **tf_activity_dataset2.csv, tf_activity_barplot_dataset2.png:** the same TF scoring applied to a second dataset, in the same format as the Q1 versions.